

Statistics & Probability



Mean, median, and mode

- 1 Find the mean and median of the data set: 12, 15, 18, 20, 25, 30.
 - 2 Find the mode and range of the data set: 7, 9, 9, 11, 14, 14, 14, 18.
 - 3 Find the mean of the data set: 8, 11, 13, 16, 22.
-

Variance and standard deviation

- 4 Find the variance and standard deviation of the data set 4, 6, 8, 10, 12, using $\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$.
 - 5 Find the variance and standard deviation of the data set 10, 12, 14, 18, 21, using $\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$.
-

Basic probability

- 6 A bag contains 5 red, 3 blue, and 2 green counters. Find the probability of drawing a blue counter.
 - 7 Using the same bag (5 red, 3 blue, 2 green counters), find the probability of NOT drawing a red counter.
 - 8 Using the same bag (5 red, 3 blue, 2 green counters), find the probability of drawing either a red or a green counter.
-

The addition and multiplication rules

- 9 Two events A and B satisfy $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Find $P(A \cup B)$.
 - 10 Two independent events A and B have $P(A) = 0.3$ and $P(B) = 0.6$. Find $P(A \cap B)$.
 - 11 Two events A and B satisfy $P(A) = 0.35$, $P(B) = 0.45$, and $P(A \cup B) = 0.65$. Find $P(A \cap B)$.
-

Conditional probability from a two-way table

- 12 A survey of 200 students found 90 study French, 70 study Spanish, and 30 study both. Find the probability a student studies Spanish, given that they study French.

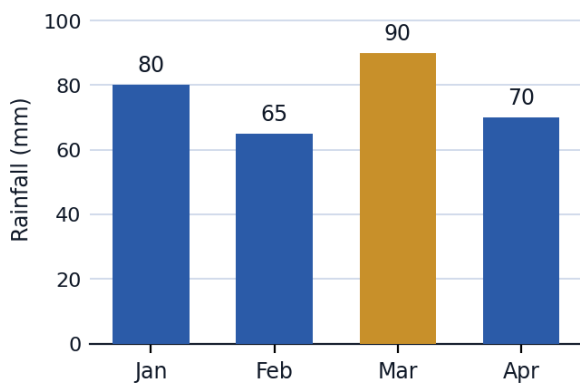
- 13 In the same survey of 200 students (90 study French, 70 study Spanish, 30 study both), find the probability that a student studies neither language.
-

Tree diagrams

- 14 A box has 4 red and 6 blue balls. Two balls are drawn without replacement. Find the probability both are red.
- 15 Using the same box (4 red, 6 blue balls, drawn without replacement), find the probability that exactly one of the two balls drawn is red.
-

Reading and interpreting a bar chart

- 16 The bar chart shows average monthly rainfall (mm) for a city over four months. Find the mean monthly rainfall, and find the percent decrease from March to April.



- 17 Using the same rainfall data (Jan 80, Feb 65, Mar 90, Apr 70 mm), find what percent of the four-month total fell in January.
-

Cumulative frequency and quartiles

- 18 A data set has lower quartile $Q_1 = 12$, median $Q_2 = 18$, and upper quartile $Q_3 = 25$. Find the interquartile range, and explain what it measures.
- 19 A data set has lower quartile $Q_1 = 15$ and upper quartile $Q_3 = 34$. An outlier boundary is defined as any value more than $1.5 \times \text{IQR}$ beyond Q_3 . Find this upper boundary.
-

A well-designed word problem: quality control

- 20** This question has two parts. A factory has two machines. Machine A produces 60% of the items and has a 2% defect rate; Machine B produces 40% of the items and has a 5% defect rate. (a) Find the overall probability that a randomly selected item is defective. (b) Given that an item is defective, find the probability it came from Machine B.